

MERN Stack Training Course

Duration: 3 Months [40 Sessions]

Course Overview

This course provides a comprehensive, hands-on training on the MERN Stack (MongoDB, Express, React, Node.js). By the end of the course, students will be able to build full-stack web applications and deploy them in production.

Course Outline

Session	Topic	Subtopics / Activities
1	Course Introduction	Overview of MERN stack, setting up development environment (VS Code, Node.js, MongoDB, Git)
2	JavaScript Fundamentals I	Variables, data types, operators, loops
3	JavaScript Fundamentals II	Functions, ES6 features (arrow functions, template literals, destructuring)
4	JavaScript DOM	Objects, arrays, array methods, DOM manipulation basics
5	Node.js Basics	Node.js introduction, event loop, modules, npm, simple server creation
6	Node.js Core Modules	fs, path, os, http modules
7	Express.js Basics	Setting up Express, routes, middleware, request-response handling
8	REST API Basics	CRUD operations, REST principles, API endpoints
9	API Testing	Using Postman, error handling, HTTP status codes
10	Express Middleware	Custom middleware, logging, CORS, environment variables
11	MongoDB Basics	NoSQL concepts, CRUD in Mongo shell
12	Mongoose ORM	Schemas, models, validation, connecting to Node.js

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Session	Topic	Subtopics / Activities
13	MongoDB Integration	Implementing CRUD APIs with Express and MongoDB
14	Advanced Mongoose	References, population, embedded documents
15	Aggregation	Aggregation pipelines, query optimization, indexing
16	Authentication Basics	JWT, bcrypt password hashing, login/signup APIs
17	React Introduction	JSX, components, props, project setup
18	React State Management	useState, event handling, forms
19	React Routing	React Router, nested routes
20	React Forms	Controlled vs uncontrolled components, validation
21	useEffect Hook	Fetching data from APIs, lifecycle effects
22	Context API	Global state management
23	Advanced Hooks	useMemo, useCallback, custom hooks
24	Styling React	CSS modules, Tailwind CSS, responsive design
25	React Advanced Concepts	Conditional rendering, lists, keys, error boundaries
26	Connecting Frontend	Axios/Fetch, connecting React frontend with Express backend
27	Authentication Flow	JWT handling in frontend, protected routes, signup/login flow
28	Route Protection	Frontend and backend route guards, auth middleware
29	File Uploads	Multer setup, uploading images/PDFs, React file preview
30	Advanced API Handling	Pagination, filtering, search
31	Deployment Basics	Heroku, Netlify, MongoDB Atlas, environment variables
32	Environment	Dev vs production, security best practices
33	Debugging	Error handling, logging, troubleshooting full-stack apps

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Session	Topic	Subtopics / Activities
34	Real-time Features	Socket.io introduction, chat module (optional)
35	Redux (Optional)	State management with Redux/Redux Toolkit
36	Project Planning	Requirements analysis, database schema design
37	Project Backend	Backend setup, API creation for project
38	Project Frontend	Connecting frontend with backend, UI setup
39	Project Implementation	Authentication, CRUD operations, final UI integration
40	Project Demo	Presentation, optimization tips, Q&A, course wrap-up